

Evaluating Diffusion Models for Cosmological Simulation Data: Memorization, Generalization, and Scientific Fidelity

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The question

Generative emulators promise **cheap surrogates** for expensive cosmological simulations — but an emulator that **memorizes** its training set produces confident nonsense when used for inference. We train diffusion models on CAMELS HI maps and ask:

1. When does the model stop **copying** training fields and start **sampling** the underlying distribution?
2. Does conditional generation actually **respond to the requested cosmology** θ ?
3. Are generated fields **statistically faithful** (one-point PDF, power spectrum)?

Data & models

- **CAMELS** (IllustrisTNG) HI fields $\rightarrow 128 \times 128$ slices; 6 parameters $\Omega_m, \sigma_8, A_{SN1}, A_{SN2}, A_{AGN1}, A_{AGN2}$.
- **Diffusion**: UNet denoisers (widths 64 / 128 / 256); conditional variant injects θ at every level via cross-attention.
- **Sampling**: 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep**: $N = 2^7 \rightarrow 2^{14}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

Detecting memorization

For each generated sample x_j , find its nearest training field in pixel, PCA, and SSCD embedding space:

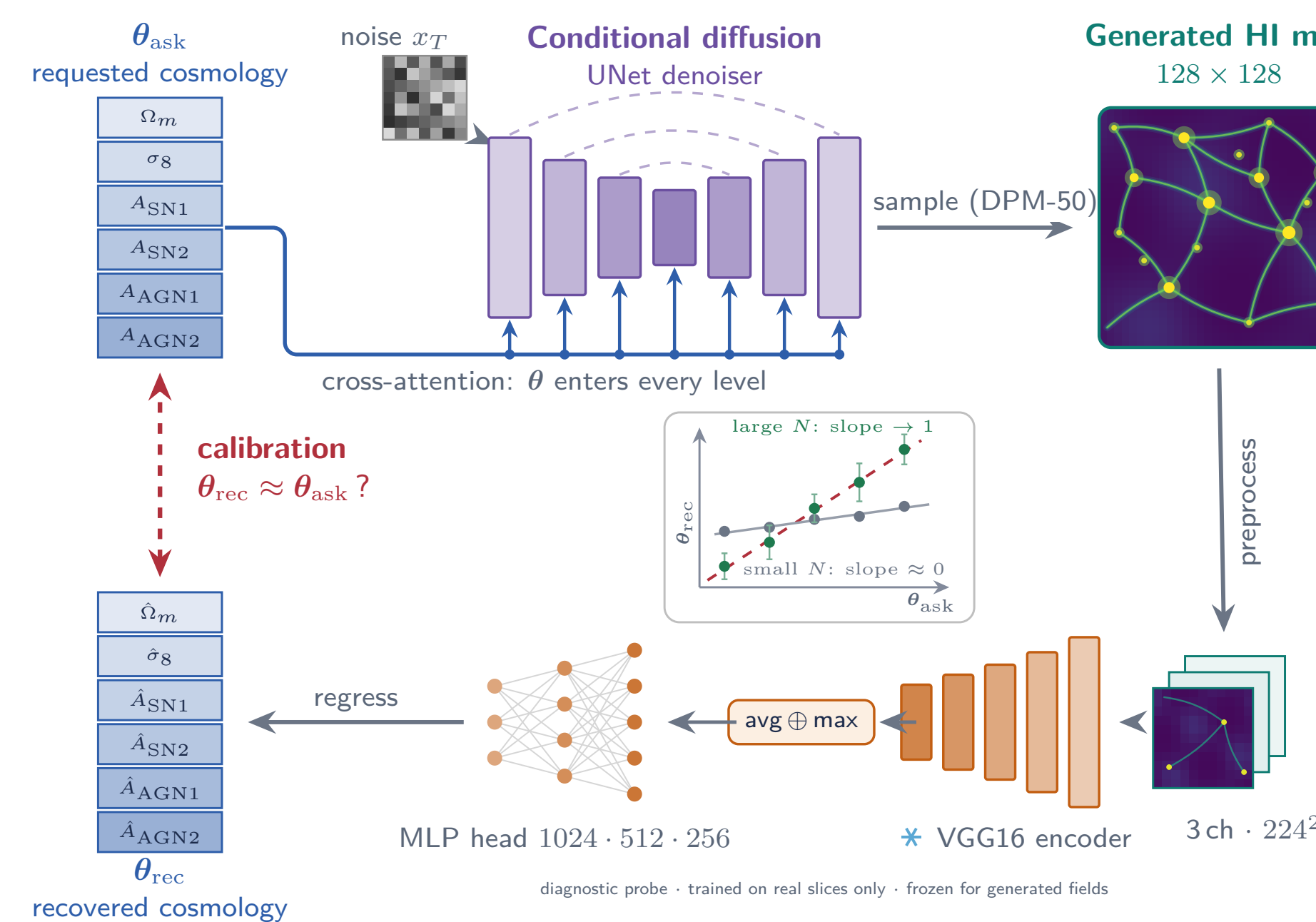
$$s_j = \max_i \text{sim}(x_j, y_i), \quad \text{GL}(\tau) = 1 - \Pr[s_j > \tau]$$

with τ chosen **adaptively** from leave-one-out train–train similarities. Independent architectures trained on the same data must agree if they learned the same distribution $\rightarrow$ **cross-model reproducibility** check.

drop in: figs/nf_generalize_fig2_u128_generated_vs_pixel_nn.png
quickcheck notebook $\rightarrow$ “Generated vs closest training slices” cell (generated / nearest train / difference)

Calibration: does the generator obey θ ?

Ask the conditional model for **held-out cosmologies**, then read the cosmology back from its generated maps with a **frozen probe**. A memorized model returns the training distribution no matter what you ask for; a generalizing model tracks the request.



Probe safeguards: VGG16 frozen (ImageNet); regression head trained on *real* slices from non-held-out simulations only; held-out sims 900–931 used exclusively for testing and calibration.

Calibration results

drop in: figs/vgg_encoder_omega_real_test_poster.png
VGG notebook $\rightarrow$ encoder sanity check on real held-out fields

drop in: figs/bias_probe_omega_m_best_vgg_poster.png
VGG notebook $\rightarrow$ recovered vs requested Ω_m bias probe

Probe skill on **real** held-out fields (R^2 , best encoder): Ω_m 0.91, σ_8 0.74, A_{SN1} 0.45, A_{SN2} 0.33, $A_{AGN1} \approx 0$, A_{AGN2} 0.10 — astrophysical feedback is only weakly imprinted on single HI slices, so calibration leads with Ω_m .

Headline: recovered-vs-requested Ω_m slope $\approx$ [X.XX] in the high-data regime vs [Y.YY] under memorization; spread across seeds *grows* with N — evidence of true conditional sampling, not regurgitation.

Memorization $\rightarrow$ generalization transition

drop in: figs/nf_generalize_fig2_poster_pca_generalization_adaptive_tau.png
quickcheck notebook $\rightarrow$ Fig. 2-style GL curves (adaptive τ , PCA/SSCD)

- **Small N** : nearest-neighbor panels show near-duplicates of training fields; $\text{GL} \approx 0$.
- **Growing N** : sharp rise of GL — the model switches from copying to producing novel fields.
- Transition location is **consistent across** PCA and SSCD features and across UNet-64 / 128 / 256.

Physical fidelity

drop in: figs/fidelity_onepoint_pk.png
quickcheck notebook $\rightarrow$ “Detailed one-point and $P(k)$ ” cell (histogram overlays + $P(k)$ ratio)

In the generalization regime, one-point PDFs overlap the real reference and mean $P(k)$ ratios stay near unity across scales — the fields are not just novel, they are **statistically right**.

Takeaways

1. Diffusion models on CAMELS show a clear **memorization** $\rightarrow$ **generalization transition** with training-set size, located consistently by NN checks, PCA/SSCD embeddings, and cross-model reproducibility.
2. Under memorization, recovered cosmology is **tight but biased** toward the training mean; in the high-data regime it **tracks** θ_{ask} with realistic sample-to-sample spread.
3. Generated fields pass **one-point and $P(k)$** checks only in the generalization regime — a practical certification recipe for simulation emulators before they are trusted for inference.

Ongoing: full 6-parameter recovery, classifier-free guidance, 3D fields; workshop paper in preparation.
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